Algorithm 1: Dynamical balancing of Distillation and CKA map loss in Eq. 3

Data: $100 \geq$ Original Accuracy $> 0; 100 \geq$ Current Validation Accuracy > 0;
Accuracy Threshold $\geq 0; 1 \geq$ Scaling Factor $> 0;$ Initial Lambda > 0
Result: λ
$\text{acc}_0 \leftarrow$ Original Accuracy;
$\text{acc}_1 \leftarrow$ Current Validation Accuracy;
$\eta \leftarrow$ Accuracy Threshold;
$\alpha \leftarrow$ Scaling Factor;
$\lambda \leftarrow$ Initial Lambda;
$\delta_{\text{acc}} \leftarrow \text{acc}_0 - \text{acc}_1$;
if $\delta_{\text{acc}} > \eta$ **then**
  $\lambda \leftarrow \lambda \times \alpha$
else
  $\lambda \leftarrow \lambda/\alpha$
end

Return λ;

B Additional Results

B.1 CKA sensitivity to invertible linear transformations

We also experiment, for linear CKA, with a type of transformation that isn't considered by our theoretical results but which we deemed interesting to analyze empirically, namely multiplications by invertible matrices. Consider a matrix $M \in \mathbb{R}^{d \times d}$ whose elements are sampled from a Gaussian with mean μ and standard deviation σ. We verify the invertibility of M since it is not guaranteed and only keep invertible matrices. We show the CKA values between X and the transformed $Y = XM$ in Fig. 9). Since this is an invertible linear transformation we would expect it to only modestly change the representations in X and the CKA value to be only slightly lower than 1. However, we observe that even for small values of μ and σ, CKA drops to 0, which would indicate that the two sets of representations are dissimilar and not linked by a simple, invertible transformation.

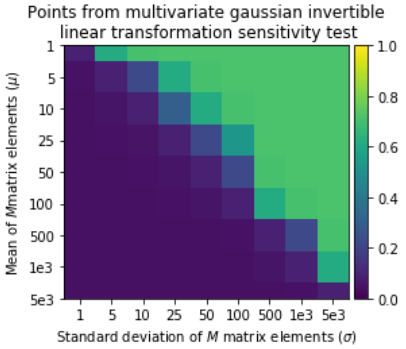

Figure 9: Linear CKA values between the artificial representations X and $Y = XM$ with M being an invertible matrix with elements sampled from $\mathcal{N}(\mu, \sigma^2)$ as a function of μ and σ. The mean and standard deviation across 10 random instantiations the translation direction and M are shown.

While Th.1 of Kornblith et al. (2019) implies that invariance to invertible linear transformations is generally not a desirable property for ANN representation similarity measures, there are relatively common scenarios in which the hypotheses of the theorem are not necessarily respected, i.e. where the dataset size is larger than the width of the layer. Such is the case in smaller ANNs or even at the last layers of large models which are often fully connected and of far smaller size than the input space or the intermediate layers. Given these situations we see no reason to completely dismiss this invariance as being possibly desirable in certain, albeit not all, contexts.

B.2 Wider Networks

In Sec. 4.3 we investigated whether the CKA of the wide networks can be mapped to a thin network (see Fig. 7) using ResNet-34 models and the CIFAR10 dataset. In Fig. 10, we use the same networks (trained on CIFAR10) and measure their CKA similarity maps under the Patch Camelyon dataset (Veeling et al., 2018). Patch Camelyon dataset contains histopathologic scans of lymph node sections, which is drastically different from the CIFAR10 dataset both in terms of pixel distribution and the semantics of the data. As we can see in Fig. 10, even under this drastic shift in data distribution

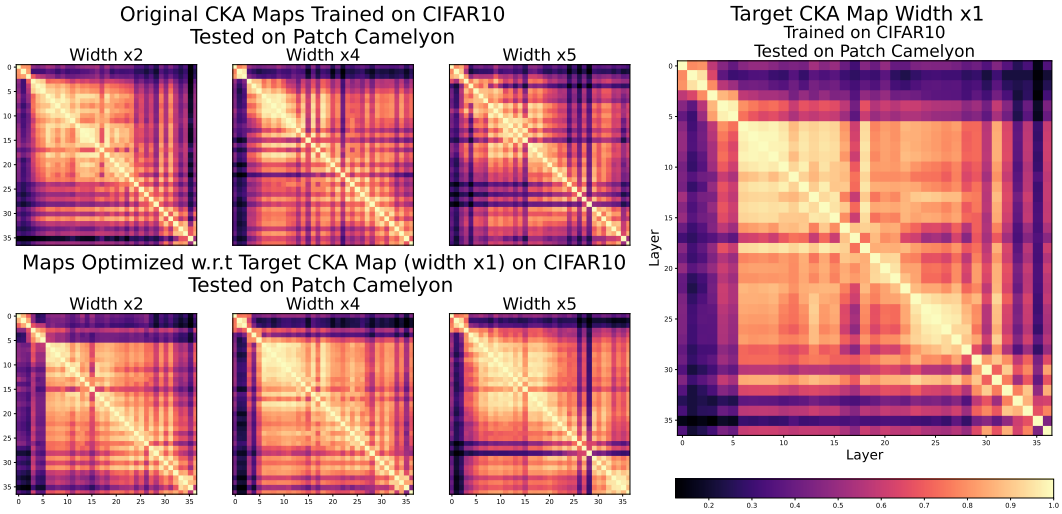

Figure 10: In Fig. 7 we are presented a series of ResNet-34 networks of different widths and their corresponding CKA Maps, which are modified to produce CKA maps of thin networks using the CIFAR10 dataset. We used these networks and measured their CKA maps using Patch Camelyon dataset. Top row **Original** shows the unaltered CKA map of the networks derived from "normal" training on CIFAR10, tested on Patch Camelyon dataset. **Optimized** shows the CKA map of the networks after their map is optimized to mimic the thin network *target* CKA map using CIFAR10, tested on Patch Camelyon dataset.

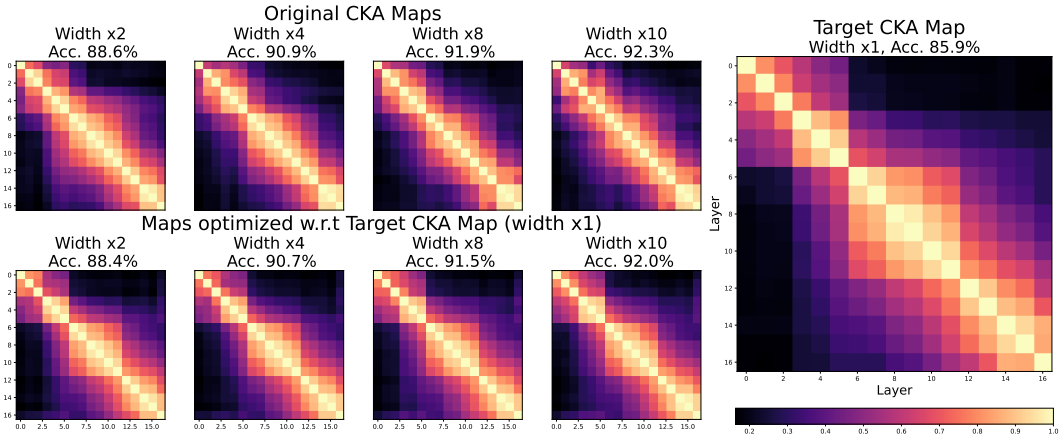

Figure 11: VGG style networks of different widths and their corresponding CKA Maps are modified to produce CKA maps of thin networks. Top row **Original** shows the unaltered CKA map of the networks derived from "normal" training. **Optimized** shows the CKA map of the networks after their map is optimized to mimic the thin net *target* CKA map.

the CKA maps of the networks *Optimized w.r.t Target CKA Map* resemble the CKA map of the thin target network, suggesting the generalizability of the CKA map optimization.

The network architecture presented in the **Wider networks** experiments seen in Sec. 4.3 is ResNet-34. We experimented with a VGG style network architecture to broaden our findings to other network architectures (see Sec. A.2 for details). As we can see in Fig. 11 we observe similar results to the ones shown in Fig. 7.

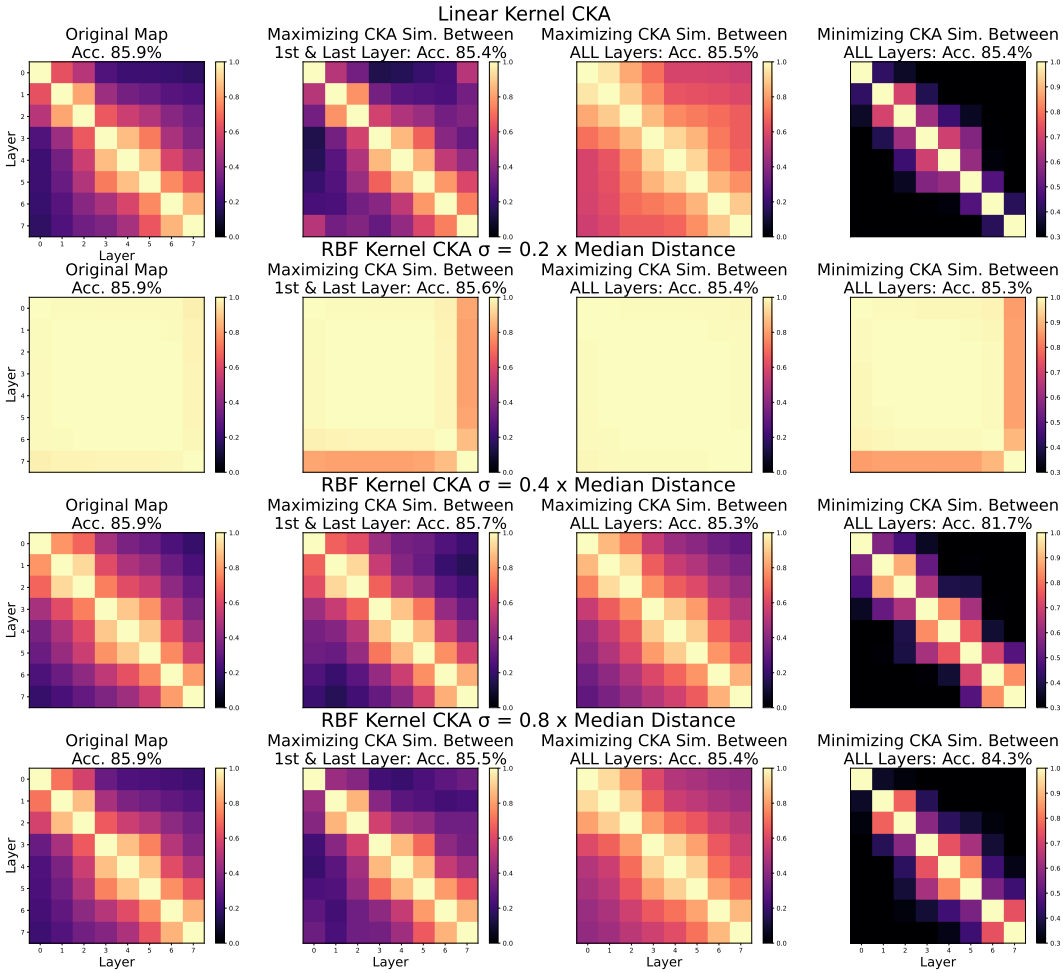

Figure 12: Original Map is the CKA map of a network trained on CIFAR10. We manipulate this network to produce CKA maps which: (1) maximizes the CKA similarity between the 1st and last layer, (2) maximizes the CKA similarity between all layers, and (3) minimizes the CKA similarity between all layers. In cases (1) and (2), the network experiences only a slight loss in performance, which counters previous findings by achieving a strong CKA similarity between early and late layers.

B.3 RBF Kernel

In Fig. 12 we extend our results shown in Fig. 5 to other bandwidth values commonly used for the RBF kernel CKA (Kornblith et al., 2019). When the CKA values are meaningful, we observe that the RBF kernel CKA values can be manipulated via the procedure described in Sec. 4.3.

B.4 CKA Map Optimization via Logistic Loss

In Sec. 4.3, we manipulated a network's CKA map, while closely maintaining its outputs via the distillation loss seen in Eq. 3. However, a logistic loss also works in this setting, i.e. the substitution of the distillation loss with cross-entropy loss in Eq. 3 yields similar results. In Fig. 13, we repeated the linear CKA experiments seen in the first row of the Fig. 5 using cross-entropy loss instead of distillation loss.